DeepSeek: Mathematical Foundations & Algorithms

How Group Relative Policy Optimization (GRPO) teaches LLMs to reason

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Introduction

- **DeepSeek** is an AI company founded by **Liang Wenfeng** on 17 July 2023 in Hangzhou, China, and funded by the hedge fund High-Flyer [1].
- **DeepSeek-R1** (January 2025) is a reasoning model that performs on par with OpenAI's o1 in maths, coding and reasoning [2].
- Open weights under the MIT licence: **DeepSeek-R1-Zero**, **DeepSeek-R1**, and six smaller distilled models based on **Qwen** and **Llama** [3].
- **Cost:** R1 is built on DeepSeek-V3, whose final training run used 2.788M H800 GPU hours on a 2,048-GPU cluster, about **$5.6M** at $2 per GPU hour. This excludes earlier research and ablation runs [4].

DeepSeek-V3: the base model behind R1

- **Mixture-of-Experts (MoE):** 671B parameters in total, but only **37B are active** for each token [4].
- **Multi-head Latent Attention (MLA):** compresses keys and values to shrink the memory needed during generation.
- **Multi-token prediction (MTP):** a *training objective* that predicts several future tokens; the extra module can also be reused for **speculative decoding** at inference.
- **Efficient training:** FP8 mixed-precision training and the DualPipe pipeline-parallel algorithm.

Key idea

Spend compute only on the experts each token needs, and train in lower precision, so a very large model becomes affordable to train.

DeepSeek-R1 versus OpenAI o1

Benchmark (metric)	**DeepSeek-R1**	**OpenAI o1-1217**
AIME 2024 (pass@1)	**79.8**	79.2
MATH-500 (pass@1)	**97.3**	96.4
Codeforces (percentile)	96.3	**96.6**
GPQA Diamond (pass@1)	71.5	**75.7**
MMLU (pass@1)	90.8	**91.8**
SWE-bench Verified (resolved)	**49.2**	48.9

Table 1: Reported scores; higher is better [3].

R1 matches o1 on maths and coding while being openly released.

What DeepSeek is addressing

- **Reasoning through reinforcement learning:** R1-Zero showed that step-by-step reasoning can emerge from RL with simple rule-based rewards, without supervised fine-tuning first [2].
- **Training efficiency:** sparse MoE activation and FP8 training cut the compute needed [4].
- **Open access:** open weights lower the barrier for developers and organisations, especially outside the largest tech companies [5].

Starting point: Proximal Policy Optimization (PPO)

For a question q and output tokens o_t, PPO maximises [6]:

$$J_{\text{PPO}}(\theta) = \mathbb{E}\Big[\min\left(r_t(\theta)A_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t\right)\Big],$$

where the probability ratio between the new and old policy is

$$r_t(\theta) = \frac{\pi_\theta(o_t \mid q, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t \mid q, o_{<t})}.$$

- A_t: advantage, normally estimated with a learned **value (critic) model**.
- The clip with ϵ stops any single update from moving the policy too far.

The problem for LLMs

The critic is usually as large as the policy itself, roughly doubling memory and compute [7].

GRPO: replace the critic with a group baseline

For each question q, sample a **group** of G answers $\{o_1, \ldots, o_G\}$ from the old policy and score them with rewards $r_1, \ldots, r_G$ (for example: is the final answer correct, is the format right) [2, 7].

Group-relative advantage (every token of answer i gets the same value):

$$\hat{A}_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}$$

- **No critic network:** the group mean is the baseline, saving memory.
- **Variance reduction:** answers are judged against their siblings, not on an absolute scale.
- Answers better than the group average are pushed up; worse ones are pushed down.

The GRPO objective

$$J_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\left\{\ell_{i,t}(\theta) - \beta\,\mathbb{D}_{\text{KL}}\left[\pi_\theta \,\|\, \pi_{\text{ref}}\right]\right\}\right],$$

with the clipped per-token term

$$\ell_{i,t}(\theta) = \min\left(r_{i,t}(\theta)\hat{A}_i,\ \text{clip}(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_i\right).$$

- The **KL penalty** (weight β) keeps the policy close to a reference model and is added to the loss directly, not folded into the reward as in PPO [7].
- KL is estimated per token with an unbiased, always-positive estimator:

$$\mathbb{D}_{\text{KL}} = \frac{\pi_{\text{ref}}}{\pi_\theta} - \log\frac{\pi_{\text{ref}}}{\pi_\theta} - 1.$$

Why it works: the policy-gradient view

The policy gradient weights each token's log-probability by its advantage:

$$\nabla_\theta J(\theta) = \mathbb{E}\Big[\sum_t \nabla_\theta \log \pi_\theta(o_t \mid q, o_{<t})\, \hat{A}_i\Big].$$

- Subtracting the group mean is a **baseline**: it leaves the expected gradient unchanged but lowers its variance.
- Dividing by the group standard deviation rescales every question to a similar size of update.

Known caveats (later work)

The length term $1/|o_i|$ and the standard-deviation scaling introduce biases toward longer wrong answers and toward very easy or very hard questions; "Dr. GRPO" removes them [8].

GRPO: the algorithm

Algorithm 1 GRPO (simplified, after [7])

Input: policy π_θ, reference π_{ref}, reward function R, group size G
for each training step **do**
    Sample a batch of questions q
    for each question q **do**
        Sample G answers $o_1, \ldots, o_G \sim \pi_{\theta_{\text{old}}}(\cdot \mid q)$
        Score them: $r_i = R(q, o_i)$
        Group-normalise: $\hat{A}_i = (r_i - \text{mean}(r))/\text{std}(r)$
    end for
    Update θ by gradient ascent on $J_{\text{GRPO}}(\theta)$
    $\theta_{\text{old}} \leftarrow \theta$
end for

From R1-Zero to R1

- **R1-Zero:** GRPO applied directly to the V3 base model with rule-based rewards (answer accuracy and output format). Reasoning emerges, but outputs mix languages and are hard to read [2].
- **R1** adds a multi-stage pipeline [2]:
  1. *Cold start:* a small set of readable long-reasoning examples for supervised fine-tuning.
  2. *Reasoning RL:* GRPO, with an extra **language-consistency reward**.
  3. *Rejection sampling + SFT:* keep good RL outputs and add general data.
  4. *RL for all scenarios:* a final round for helpfulness and harmlessness.
- **Distillation:** R1's outputs train smaller Qwen and Llama models [3].

Potential applications

- **Natural language processing:** generation, translation, sentiment analysis.
- **Scientific research:** step-by-step reasoning for maths and science problems.
- **Healthcare:** supporting analysis of biomedical text and data, with human oversight.
- **Finance:** risk assessment and analysis, where lower inference cost matters [5].

Market reaction and implications

Market reaction (27 January 2025)

- Nvidia fell nearly 17%, losing about **$589 billion** in market value, the largest one-day loss in US history; the Nasdaq Composite fell 3.1% [9].

Implications

- Low-cost, open models put pressure on closed providers to cut prices [5].
- Wider access to strong models, alongside open questions about data security and governance [1].

Outlook

- **Practical AI:** reasoning models moving into healthcare, education and business workflows.
- **Cheaper, open models:** efficiency gains make strong models available beyond large labs.
- **Open research questions:** better reward design, removing GRPO's biases, and making reasoning models calibrated and trustworthy.

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